

Chapter 4

International Trade

International trade, although often viewed through the lens of contemporary globalization, has existed for millennia. Long before the modern global economy emerged, goods traveled vast distances along networks such as the ancient Silk Road, connecting distant cultures and economies. Historically, trade has often been an engine of prosperity, fueling the rise of states like Sriwijaya (in modern-day Indonesia), the Republic of Venice, and the British Empire. Trade also has political effects. It sometimes led nations into conflict—as with the Opium Wars—and at other times fostered peaceful cooperation, as envisioned by Montesquieu in his idea of *gentle commerce* (in French, *doux commerce*)? This chapter seeks to explore precisely such questions: How does international trade create economic benefits? Who are the winners and losers of opening borders? And ultimately, under what circumstances might free trade be desirable or undesirable?

Throughout the last two centuries, international trade has undergone dramatic cycles of expansion and contraction. The late 19th century witnessed a rapid acceleration in global commerce (now called the *first globalization*) driven by advances in transportation technology, relative peace following the Napoleonic Wars, and the growth of colonial empires (see Figure 4.1). However, this expansion was sharply interrupted by the turmoil of two World Wars and the resurgence of protectionist policies during the Great Depression of the 1930s. This illustrates how political and economic instability can quickly reverse economic integration. Trade recovered robustly after World War II, supported by new international institutions such as the General Agreement on Tariffs and Trade (GATT)—today the World Trade Organization (WTO)—which facilitated the systematic removal of tariff barriers among industrialized nations

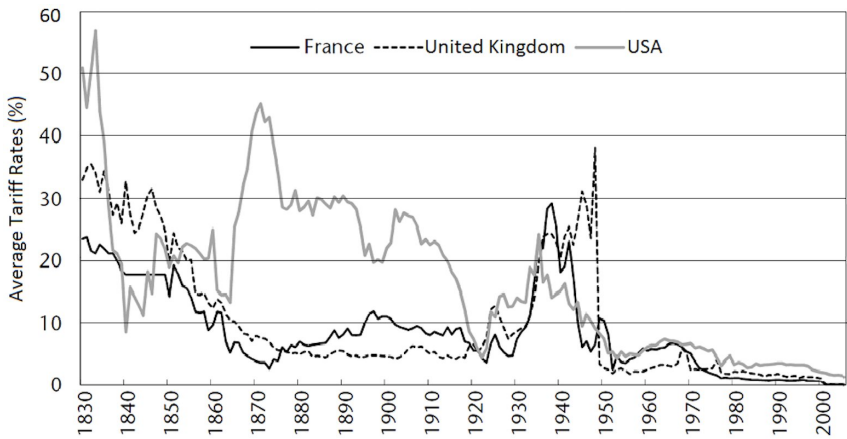
Figure 4.1: Share of exports in global production



Source : Fouquin M. et Hugot J. (2016), “Two Centuries of Bilateral Trade and Gravity Data : 1827-2014”, CEPII working paper.

(see Figure 4.2). This renewed openness paved the way for the *second globalization*, which began in the late 1980s with the collapse of the Eastern bloc and China’s progressive integration into global markets.

Figure 4.2: Average tariff rates on total imports



Source : Fouquin M. et Hugot J. (2016), “Two Centuries of Bilateral Trade and Gravity Data : 1827-2014”, CEPII working paper.

These historical developments prompt a key question: is free trade desirable? Surveys conducted between 2010 and 2012 indicate that only

46% of Americans viewed positively the 1994 free trade agreement between Canada, Mexico, and the United States (NAFTA—replaced in 2018 by the United States–Mexico–Canada Agreement, USMCA), despite 95% of economists agreeing that the agreement was beneficial for Americans (Sapienza et Zingales, 2013).

In this chapter, we will study in detail the Ricardian model, which forms the foundation of international trade theory and highlights the gains associated with international trade. To understand certain objections to free trade, we will then go beyond the Ricardian model to examine who are the winners and losers from trade. Finally, we will review other effects of free trade that may—or may not—justify the opening of borders to trade.

4.1 The Ricardian Model

The mathematician Stanisław Ulam once challenged the renowned economist Paul Samuelson (1969 Nobel laureate) to identify a truly significant yet non-obvious discovery in the social sciences. Samuelson eventually answered with the concept of *comparative advantage*, describing it as “logically true [...] yet non-trivial, as shown by the thousands of important and intelligent people who, even after it was explained, could neither grasp nor believe it.”

Why did Samuelson single out this particular idea as both surprising and essential? In this section, we will uncover the reasoning behind comparative advantage, a concept first introduced by the British economist David Ricardo in 1817. What makes this insight so compelling—and yet so difficult for many to accept? Why is it that, according to Ricardo’s theory, countries always benefit from trading with one another, even when one country appears less productive in every sector?

4.1.1 Productivity, Absolute Advantage and Comparative Advantage

One of the main reasons countries trade with each other is that they differ significantly in their productivity across various economic sectors. For example, France can produce cars more efficiently than Madagascar, while Madagascar is more efficient at producing vanilla. To carefully analyze the consequences of these productivity differences, the Ricardian model focuses on a simplified scenario: two countries producing two goods, using only one factor of production—labor.

Consider a simple scenario with only two countries, France and Germany, each producing only two goods: boats and cars. Suppose producing

one boat requires 100 hours of labor in Germany, compared to 125 hours in France, and producing one car takes 50 hours in Germany versus 125 hours in France (see Table 4.1). Clearly, Germany is more productive in both industries. At first glance, this might suggest that opening trade would harm France, as German producers could potentially dominate the market with lower-priced products. But is this necessarily the case? What really happens when the borders between the two countries are opened?

To analyze the question systematically, suppose that each country is endowed with 500 hours of labor. Under these circumstances:

- Germany could produce either 5 boats and no cars, or 10 cars and no boats.
- France could produce either 4 boats and no cars, or 4 cars and no boats.

Tableau 4.1: Number of labor hours required to build a boat and a car in each country

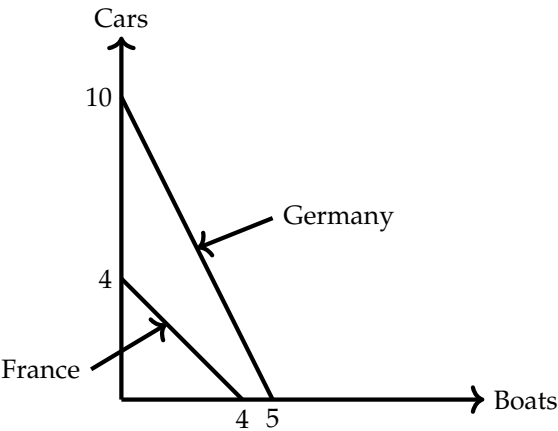
	Boat	Car
Germany	100	50
France	125	125

Moreover, each country can choose any combination between these extremes. For instance, France might opt to produce 2 boats and 2 cars. We represent all possible combinations of goods each country can produce with its available resources as a **production frontier**, illustrated in Figure 4.3. Germany’s absolute advantage implies its production frontier lies entirely above France’s.

The fact that Germany is more productive than France in all sectors implies that Germany’s production frontier lies above that of France. Note that in autarky, when the economy is closed and international trade is impossible, the production frontier is also, for each country, the consumption frontier—that is, each country cannot consume a bundle of goods that exceeds its consumption possibilities.

Notice two key insights from the shape of these production frontiers. First, both are downward sloping, reflecting that producing more of one good inevitably means producing fewer of the other: there is always a trade-off. Second, the steepness of these frontiers—their slopes—captures the **opportunity cost** of producing one good in terms of the other. For example, Germany’s production frontier indicates that producing one additional boat (requiring 100 labor hours) means giving up two cars (since each car takes only 50 hours). Thus, Germany’s opportunity cost of

Figure 4.3: Production Frontiers of Germany and France



producing a boat is two cars. In France, however, making one extra boat (125 labor hours) requires forgoing just one car (also 125 hours). Despite Germany’s higher absolute productivity, France has a *lower opportunity cost* of boat production. Table 4.2 summarizes these opportunity costs.

Tableau 4.2: Opportunity Costs in Germany and France

	Boat	Car
Germany	2	1/2
France	1	1

The concept of opportunity cost is fundamental in economics and extends far beyond Ricardo’s model. In general, an opportunity cost is what must be sacrificed to obtain something else. For instance, the opportunity cost of enjoying an hour of leisure is the wage you could have earned by working during that hour—showing that from an economic perspective, leisure is not actually “free.” Similarly, a student deciding between studying economics or physics faces a clear opportunity cost: choosing economics means giving up the chance to specialize in physics. Any other option, such as computer science, is not relevant if it’s less preferred than these two alternatives.

Applying the idea of opportunity cost directly to international trade allows us to formalize Samuelson’s earlier insight as follows:

A country has a **comparative advantage** over another country in the production of a good when its opportunity cost of producing that good is lower.

It is crucial at this point to clearly distinguish the concept of comparative advantage from that of absolute advantage:

A country has an **absolute advantage** in producing a good if it can produce more of that good than another country using the same amount of resources.

In our previous example, Germany clearly holds an absolute advantage in producing both boats and cars—it is simply more productive than France in both sectors. However, comparative advantage highlights something less obvious and more subtle: despite being less productive overall, France holds a *comparative advantage* in boat production because its opportunity cost (1 car per boat) is lower than Germany’s (2 cars per boat). Conversely, Germany enjoys a comparative advantage in car production, as producing a car means giving up just half a boat, while in France it would require sacrificing an entire boat (see Table 4.3).

Tableau 4.3: Absolute and Comparative Advantages in Germany and France

	Boat	Car
Absolute Advantage	Germany	Germany
Comparative Advantage	France	Germany

In summary, an absolute advantage reflects differences in *overall* productivity levels, whereas a comparative advantage arises from *relative* productivity differences across sectors. Compared to Germany, France is relatively better at producing boats than cars, while Germany exhibits the opposite relative productivity pattern. Consequently, even if a country enjoys an absolute advantage in every sector—as Germany does in our example—it cannot hold a comparative advantage in all sectors simultaneously. Being relatively more productive in one industry necessarily implies being relatively less productive in another. Understanding this fundamental point is critical, as it sets the stage for Ricardo’s central insight: that differences in comparative advantage form the basis for mutually beneficial international trade.

4.1.2 Specialization and Trade

Autarky Situation. How exactly do the productivity differences and comparative advantages identified earlier influence trade between France and Germany? To answer this question, we start by examining a situation called *autarky*, where countries do not engage in international trade. In autarky, each country's production fully determines its consumption possibilities, and the prices of goods simply reflect domestic production costs.

Let us express the price of boats in each country. Since the only other good is cars, the price of a boat only makes sense if it is expressed relative to the price of a car. We thus define the relative price as:

$$p = \frac{P_{\text{Boat}}}{P_{\text{Car}}},$$

which amounts to normalizing the price of cars to 1 and examining the price p of boats.¹

When a country is in autarky, the price of each good simply reflects its production cost. Thus, the relative price of a boat (compared to a car) directly equals the opportunity cost of producing that boat, measured in terms of the cars forgone. For instance, because Germany requires twice as many labor hours to produce a boat as a car, the relative price of a boat in Germany naturally equals 2:

$$p^{\text{Germany}} = \frac{P_{\text{Boat}}^{\text{Germany}}}{P_{\text{Car}}^{\text{Germany}}} = 2,$$

where p^{Germany} is the price of a boat in Germany (relative to a car) under autarky. Similarly,

$$p^{\text{France}} = \frac{P_{\text{Boat}}^{\text{France}}}{P_{\text{Car}}^{\text{France}}} = 1,$$

where p^{France} is the price of a boat in France under autarky.

Uniqueness of Relative Price. What happens when borders are opened and France and Germany engage in free trade? The first important conse-

¹In the previous chapter, we focused on a single market and expressed prices in euros, under the assumption that euros can be used on other markets that we set aside. This was what we call a *partial equilibrium* analysis. We now consider all the economy's markets at once, in what is known as a *general equilibrium* analysis. In a general equilibrium context, only relative prices are defined. Thus, one can designate a particular good, such as cars, as the numeraire (the reference) and focus on the price of other goods (in this case, boats). The analysis would be unchanged if we chose boats as the numeraire.

quence is that the relative price of boats will adjust, eventually becoming identical in both countries.

To understand why, suppose momentarily that prices remain at their original autarky values, with $p^{\text{France}} = 1$ and $p^{\text{Germany}} = 2$. Under these conditions, a French individual who owns one boat could travel to Germany, trade it for two cars, then return to France and exchange those two cars for two boats. By repeating this cycle, they could obtain four boats, then eight boats, and so forth. With minimal transaction costs, this arbitrage process would lead to unlimited profits without any production. Clearly, such a scenario cannot last. In Germany, boat owners would quickly seize the opportunity to trade boats for relatively cheaper cars, causing the relative price of cars to rise and thus lowering p^{Germany} . Conversely, in France, individuals would seek to exchange cars for relatively cheaper boats, causing the relative price of boats (p^{France}) to rise.

The only situation in which no arbitrage opportunities remain is an equilibrium where relative prices equalize across countries:

$$p^{\text{France}} = p^{\text{Germany}} = p.$$

This outcome directly reflects the **law of one price**, introduced in the previous chapter, which states that in an integrated market, identical goods must trade at the same price everywhere. Once trade barriers between France and Germany are removed, boats and cars effectively trade in one unified Franco-German market. However, it is important to note that the law of one price holds strictly only in the absence of transaction costs.

On Transaction Costs

The assumption of no transportation costs might initially seem unrealistic. However, advances in logistics and the standardization of shipping containers have significantly reduced international transportation costs. Today, shipping a 63 cubic meters container from New York to Marseille costs around \$2,000, and since such a container can carry over 20 tonnes of goods, transportation costs typically represent a small fraction of a product's value. Nevertheless, trade is not frictionless for all goods. Some products, like fresh produce, must be transported quickly by air, making transportation more costly. Many services, such as healthcare, restaurants, or hairdressing, are inherently difficult to trade internationally. Differ-

ences in language also pose substantial barriers to trade. Finally, for non-standardized goods (such as textiles or toys, compared to commodities like oil or wheat), initiating trade often requires building trust relationships between exporters and importers—relationships that can be costly and time-consuming to develop. The study of these frictions has become a major area of research in international trade.

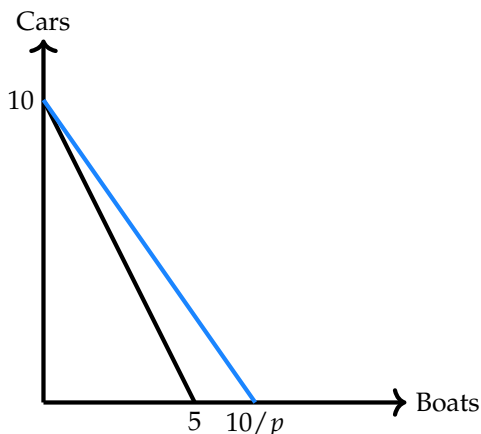
Specialization and Gains from Trade. Suppose that after opening the borders, the relative price of a boat settles at some level between the two autarky prices observed earlier—that is, p lies between 1 and 2. Thus, for now, we treat p as fixed. Later, we will explicitly examine how supply and demand interactions determine this equilibrium price. Given this assumption, how would each country respond, and what would their resulting patterns of specialization look like?

Consider Germany first. Recall that 100 hours of labor allow Germany to produce either one boat or two cars. However with $p < 2$, the relative price of a boat is lower than its autarky price (equal to 2), meaning boats have become cheaper relative to cars. Germany therefore finds it advantageous to specialize entirely in car production. By devoting all 500 hours of labor to producing cars, Germany produces 10 cars. Each car can then be exported to France in exchange for $1/p$ boats. Since $1/p$ is larger than $1/2$, Germany receives more than half a boat per car exported, thus obtaining more than the 5 boats it could have produced domestically without trade. Specifically, by exporting all 10 cars, Germany can import $10/p > 5$ boats.

This outcome is illustrated in Figure 4.4. The black line represents Germany's production frontier, identical to the one previously shown in Figure 4.3. Under autarky, this production frontier also serves as its consumption frontier. The blue line depicts Germany's consumption possibilities when trade is allowed and $p < 2$. With international trade, Germany can now consume combinations of boats and cars that exceed what would have been possible in a closed economy. This outcome clearly illustrates how specialization leads to gains from trade for Germany.

How does France respond to this new trade opportunity? Recall that in France, producing a boat requires the same labor input as producing a car. However, if the relative price is larger than 1 ($p > 1$), boats become more valuable than cars. France therefore specializes entirely in boat production, producing 4 boats. Each boat can then be exported to Germany in exchange for p cars, with $p > 1$. Thus, by exporting all 4

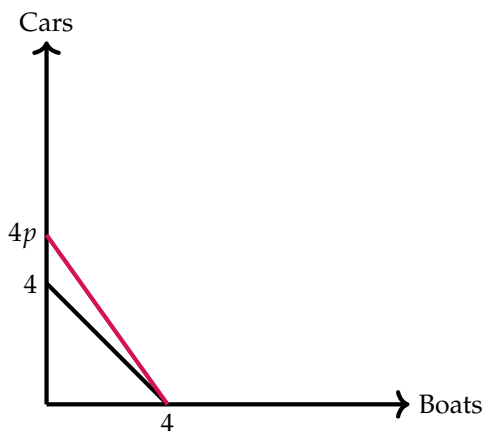
Figure 4.4: Consumption frontiers in Germany in a closed economy (black) and an open economy (blue) when $p < 2$



boats, France can import $4p > 4$ cars, which is more than the country could have produced domestically.

This scenario is illustrated in Figure 4.5. The black line indicates France's production frontier, which under autarky also served as its consumption frontier. The red line represents France's new consumption possibilities after opening to trade. Clearly, France's consumption possibilities expand in an open economy. Therefore, as soon as France engages in trade and consumes cars imported from Germany, it also benefits from free trade.

Figure 4.5: Consumption frontiers in France in a closed economy (black) and in an open economy (red) when $p > 1$



We have thus seen in this example that each country specializes in the sector where it holds a comparative advantage. This specialization makes mutually beneficial trade possible, allowing both nations to consume beyond what their own production alone would permit. Closing borders in this scenario would therefore be Pareto-inefficient, as it would leave both countries worse-off.

The precise value of the relative price p determines how the gains from trade are shared. When p is high (close to 2), the French can purchase more cars for each boat exported, thus benefiting significantly from opening to trade. Conversely, when p is low (close to 1), Germans gain relatively more from trade. Indeed, the further the equilibrium price deviates from a country's autarky price, the greater the gains that country realizes from opening its borders.

In the limiting case where $p = 1$, France no longer has a strict incentive to specialize exclusively in boat production. Whether it produces 0 or 4 boats, its consumption possibilities remain unchanged, as trade does not alter its relative prices. Consequently, France gains nothing from free trade in this special case. In a similar fashion, when $p = 2$, Germany loses its incentive to specialize and receives no benefit from trade, as its domestic price remains unchanged by the opening of trade.

Determination of the Single Price. The previous analysis assumed a fixed relative price and focused solely on the supply response of each country. To obtain a complete understanding of trade equilibrium, we now explicitly show how the equilibrium price is determined by the interaction of supply and demand.

Let us first summarize and complete the analysis of supply of boats and cars. As previously established, for all values of p strictly between 1 and 2, Germany produces 10 cars, and France produces 4 boats. Thus, the total production of the combined Franco-German economy remains unchanged as long as p lies between 1 and 2. At the boundary $p = 1$, Germany still specializes fully in car production, but France no longer necessarily specializes. Hence, total production consists of exactly 10 cars and at most 4 boats. Conversely, at $p = 2$, France fully specializes in boat production, while Germany does not necessarily specialize. Therefore, global production includes exactly 4 boats and at most 10 cars. Moreover, when $p < 1$, neither country produces boats, resulting in a total production of 14 cars; and conversely, when $p > 2$, neither country produces cars, yielding 9 boats in total.

Now that we have determined, at any price, the supply of boats and cars, we can construct the relative supply curve of boats, that is, the ratio

$$S(p) = \frac{S_{\text{Boat}}(p)}{S_{\text{Car}}(p)},$$

where $S_{\text{Boat}}(p)$ is the total supply of boats and $S_{\text{Car}}(p)$ the total supply of cars. This ratio, presented in the last row of Table 4.4, summarizes the economy's relative production across various price levels.

Tableau 4.4: Relative supply of boats as a function of the relative price.

	$p < 1$	$p = 1$	$1 < p < 2$	$p = 2$	$p > 2$
$S_{\text{Boat}}^{\text{France}}$	0	0 to 4	4	4	4
$S_{\text{Boat}}^{\text{Germany}}$	0	0	0	0 to 5	5
$S_{\text{Car}}^{\text{France}}$	4	0 to 4	0	0	0
$S_{\text{Car}}^{\text{Germany}}$	10	10	10	0 to 10	0
$S(p)$	0	0 to $\frac{4}{10}$	$\frac{4}{10}$	at least $\frac{4}{10}$	$+\infty$

Ranges indicated in the table: for $p = 1$, the relative supply $S(p)$ lies between 0 and $\frac{4}{10}$; for $p = 2$, $S(p)$ is at least $\frac{4}{10}$.

But how is the open-economy equilibrium price p determined? To answer this question, we must now explicitly introduce consumers' demand and examine how market equilibrium arises from the intersection of supply and demand.

Therefore, to determine the equilibrium price, we now move to the second step: the construction of the demand curve. Let $D_{\text{Boat}}(p)$ denote the Franco-German demand for boats as a function of their relative price p , and $D_{\text{Car}}(p)$ denote the corresponding demand for cars. The relative demand for boats is thus defined as:

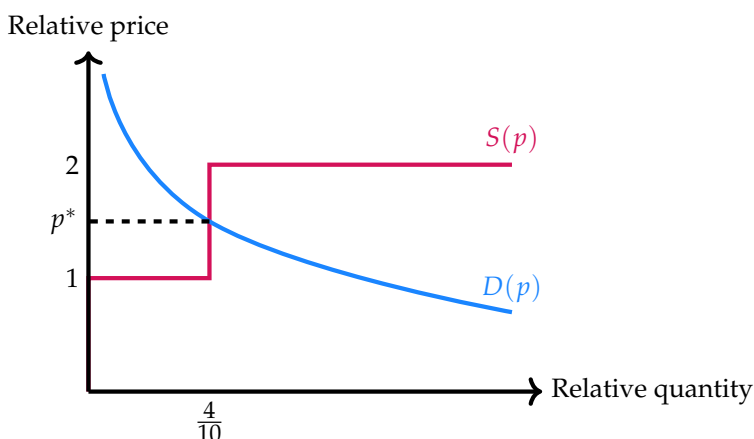
$$D(p) = \frac{D_{\text{Boat}}(p)}{D_{\text{Car}}(p)}.$$

Naturally, the demand for boats, $D_{\text{Boat}}(p)$, decreases as the relative price p increases, while the demand for cars, $D_{\text{Car}}(p)$, increases with p . Consequently, the relative demand $D(p)$ is a decreasing function of p .

Figure 4.6 illustrates both the relative supply and relative demand curves for boats. The intersection of these curves determines the equilibrium price p^* and the corresponding equilibrium relative quantity of

boats. As long as the equilibrium price lies strictly between 1 and 2, each country fully specializes according to its comparative advantage.²

Figure 4.6: Relative supply and demand of boats, and the relative equilibrium price



4.1.3 A Historical Example: Japan's Opening to Trade

In 1639, Japan effectively entered a state of autarky: with very few exceptions, the Japanese were forbidden from having any contact, including trade, with the outside world. This situation changed dramatically in 1859, when Western powers compelled Japan to open up to international commerce with relatively low customs barriers. Within a few years, Japan experienced one of the most abrupt trade shocks in history, moving from near-total isolation to almost complete openness. This historical episode provides a unique opportunity to empirically test the theory of comparative advantage.

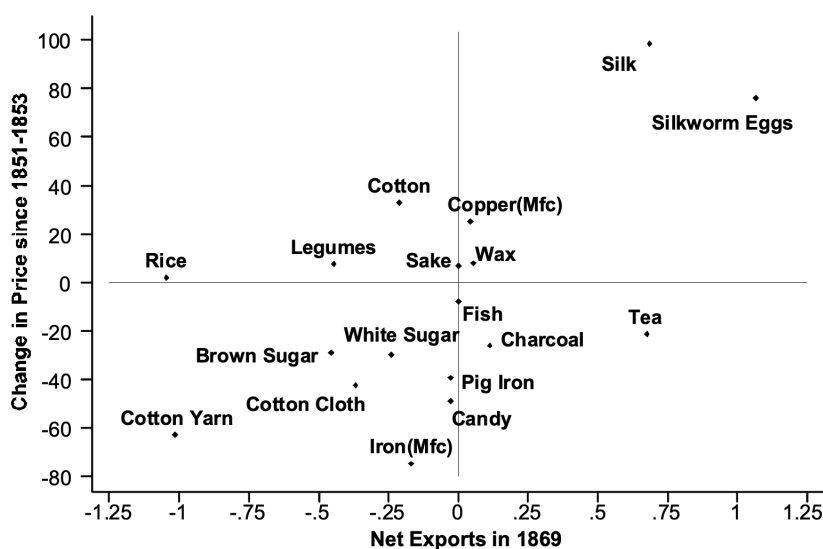
If the concept of comparative advantage is empirically valid, we should observe a clear positive correlation between Japan's net exports of particular goods and the rise in their relative prices following trade liberalization. Indeed, Ricardo's theory implies that upon opening bor-

²Why focus explicitly on *relative* supply and demand rather than treating each market separately? Because consumers choose only between cars and boats, specifying their demand for boats at a given relative price automatically determines their demand for cars. The same reasoning applies to producers' supply. Thus, equilibrium in the boat market necessarily implies equilibrium in the car market, making it incorrect to analyze these markets independently. This result stems from *Walras's law* (a key principle in general equilibrium theory), which states that in an economy with any number of goods, equilibrium in all goods but ones markets ensures equilibrium in the last good. Walras's law explains why only relative prices are meaningful in general equilibrium analysis, leading us to normalize the price of cars to 1.

ders, prices rise in sectors where domestic production costs are relatively low compared to the rest of the world—in other words, precisely where a country possesses a comparative advantage. These are the goods Japan would logically export, since specialization becomes most profitable in sectors whose prices increase significantly relative to production costs.

Bernhofen et Brown (2004) used historical archives to track the evolution of prices and trade in various goods at the time Japan opened to international commerce.

Figure 4.7: Japan's Opening to Trade



Notes: The horizontal axis indicates Japan's net exports of various goods after trade liberalization (negative values correspond to imports), while the vertical axis shows the percentage change in the relative prices of these goods over the same period. Source: Bernhofen et Brown (2004).

The findings of Bernhofen et Brown (2004) regarding changes in trade flows and goods prices are summarized in Figure 4.7. The results clearly confirm the positive correlation predicted by the theory of comparative advantage. For instance, after opening to trade, Japan became a major exporter of silk, whose price increased significantly, and an importer of cotton yarn, whose price decreased.

4.1.4 Insights from the Ricardo Model

The Ricardo model offers a wealth of fundamental insights into international trade and, more broadly, into the workings of market economies.

4.1.5 Insights from the Ricardo Model

Competitiveness. Several important insights from Ricardo's model stem directly from its emphasis on relative prices rather than absolute prices.

The first of these insights is that international trade benefits all countries involved, regardless of their absolute productivity levels. When borders open, each country expands production in sectors where it faces relatively lower costs compared to other countries. Producers in these sectors thus gain access to foreign markets characterized by strong demand and favorable relative prices. Simultaneously, consumers benefit from the opportunity to purchase goods for which the domestic production is at a relative competitive disadvantage at lower prices through imports. Because every country has at least one sector where its relative production costs are comparatively low and one where its relative production costs are comparatively high, trade creates mutual gains for producers and consumers in all trading partners.

A second, closely related insight is that the notion of "competitiveness" cannot meaningfully apply to an entire country. Competitiveness is a useful concept when analyzing individual firms: a firm that is not competitive can disappear due to stronger rivals. It also applies meaningfully at the industry level. In our previous example, opening trade eliminates the French car industry, as it is less competitive than its German counterpart. In general, the gains from international trade come from national specialization, and specialization necessarily involves shifting resources away from less competitive industries.

However, competitiveness does not extend meaningfully to the entire economy. Workers in industries that disappear, such as the French car industry in our example, do not vanish from the economy. Instead, they move to sectors offering higher wages, such as shipbuilding, reflecting a reallocation of economic resources rather than their loss.

Technological Innovation. For a country, the opening of international trade has effects analogous to those of a technological innovation. In our previous example, from France's perspective, trade liberalization can be seen as equivalent to inventing a machine that transforms boats into cars. Specifically, when $p > 1$, exporting one boat yields more than one car, and when $p < 1$, exporting one car yields more than one boat. Either scenario benefits France, since it specializes in producing the good that this new "technology" effectively multiplies. Thus, opposition to free trade sometimes mirrors opposition to technological change—similar to the Luddites in early 19th-century England, who destroyed weaving machines in reaction to the Industrial Revolution.

This analogy helps highlight the similarity between technological progress and international trade in their economic impacts. For instance, the decline in textile-sector employment in Europe resulted partly from technological innovations, which replaced manual laborers with machines, and partly from trade with China. Indeed, technological advances and the opening of trade often lead to remarkably similar outcomes for producers and workers.

Intranational Trade. The concept of comparative advantage applies not only to trade between nations but also helps explain trade within countries and even specialization within individual firms. For example, consider a company that must perform two types of tasks: managerial and administrative. Two types of employees, managers and assistants, carry out these tasks. Their respective productivities are shown in Table 4.5.

Tableau 4.5: Number of labor hours required for different tasks

	Managerial Task	Administrative Task
Manager	4	5
Assistant	20	10

In this example, the manager holds an absolute advantage over the assistant in both types of tasks, but has a comparative advantage only in managerial tasks.³ It would therefore be inefficient for the manager to undertake administrative tasks unless the assistant, specialized in administrative work, is already fully occupied. The market economy naturally encourages each worker to specialize according to comparative advantage. Such specialization leads to a Pareto-efficient outcome.

Although comparative advantage is usually presented in the context of international trade, it fundamentally describes how any market economy operates. This insight implies that market economies are inherently inclusive: everyone can contribute productively, regardless of their absolute productivity levels in different sectors. Nevertheless, in our example, the manager receives a higher salary than the assistant. The market mechanism is thus efficient, but not necessarily equitable. This efficiency-equity trade-off raises important questions regarding appropriate redistribution policies.

³This example is intentionally simplified. In practice, one can certainly find assistants who might outperform their managers in all roles including managerial ones.

4.1.6 Methodological Remarks

Is the Ricardo model—with just two countries and two goods—realistic? Clearly, it is not! Actual international trade is infinitely more complex than the simplified depiction we have considered. To approach realism, the model would need to incorporate numerous additional elements: many countries and goods, differences in product characteristics across countries, distinctions between skilled and unskilled labor, accumulated sector-specific expertise, and various frictions such as transportation costs or banking fees.

Yet, even if we incorporated all these heterogeneous dimensions, the resulting model would still be far simpler than the real world it tries to describe. Moreover, such a highly detailed model would become excessively complicated to solve and consequently as opaque as the reality we are trying to understand. Thus, a valuable economic model does not derive its usefulness from realism, but rather from the clarity and insight it provides. Even in a substantially more complex model, the fundamental mechanism of comparative advantage would still operate and shape international trade flows. The virtue of the Ricardo model is precisely that it isolates this key mechanism, enabling us to understand its logic and significance with precision and clarity.

The essential question is not whether the model accurately depicts reality, but whether its central insights remain valid when we introduce additional forms of heterogeneity we consider relevant. In other words, which specific assumptions must we modify to undermine the logic of comparative advantage? This form of reasoning characterizes scientific inquiry in economics. It represents an approach to mathematical modeling that is more conceptual and less purely descriptive than what is typically found in natural sciences such as physics.

Economics constantly navigates between the complexity of reality and the abstraction of mathematical models. It would certainly be dangerous to confuse the simplified representation provided by a model with the reality itself. Nonetheless, simplified models can deliver crucial insights. This interplay between abstraction and reality constitutes the subtlety and distinctive strength of economics.

Ricardo vs. the *Corn Laws*

In the 18th century, England's early Industrial Revolution allowed the country to grow wealthy by importing agricultural goods in exchange for its manufactured products. In the early 19th century, Napoleon's continental blockade severely restricted this trade, sharply increasing cereal prices in Britain. After Napoleon's defeat in 1815, trade should have resumed, reducing prices and benefiting consumers—but harming British landowners. Under the influence of these landowners, Parliament enacted the famous *Corn Laws* in 1815, imposing heavy restrictions on cereal imports.

It was against this backdrop that the British economist, financier, and Member of Parliament David Ricardo developed his theory of comparative advantage in 1817. Ricardo used his theory explicitly to argue in favor of free trade and against the *Corn Laws*. Although Ricardo died in 1823, his influential analysis significantly contributed to the eventual repeal of the *Corn Laws* in 1846, a turning point in British political history.

The case of the *Corn Laws* illustrates that free trade could negatively affect certain groups, such as landowners. To fully understand why trade can create winners and losers, we must move beyond Ricardo's simple framework.

4.2 Winners and Losers from Free Trade

If free trade is so beneficial, why does it often encounter fierce opposition? It could be that the logic of comparative advantage is simply difficult to grasp, generating unfounded fears. A more plausible reason is that free trade does not benefit everyone equally. In the Ricardo model previously considered, all consumers within a country are identical, and all producers have similar characteristics. However, when we allow for significant differences among workers, it becomes clear that free trade can harm some groups. For instance, a French worker specialized in automobile manufacturing might be severely disadvantaged by trade liberalization if their productivity in shipbuilding is much lower.

More advanced international trade models can incorporate heterogeneity among consumers and producers; however, these models tend to be cumbersome and difficult to analyze. To clearly understand who benefits and who suffers from free trade, we simplify our analysis by focusing on the special case of a small open economy. An economy is

defined as “small” if opening its borders to trade does not affect global prices.

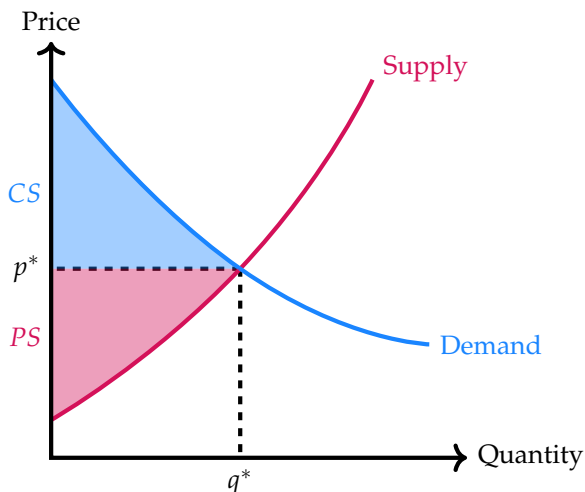
For example, Belgium can be viewed as a small open economy in the oil market: whether Belgium imports slightly more or less oil has negligible influence on global oil prices. By contrast, when the United States significantly changes its oil imports or exports, the impact on world prices can be substantial. Thus, the assumption of a small open economy would not be appropriate in that case.

4.2.1 Opening the Borders of a Small Economy

In the previous chapters, we introduced the notion of surplus and described how it is allocated between consumers and producers. Let us now apply this analysis to the context of international trade.

Consider a small economy and one specific market within it—such as the steel market. Before opening to international trade, the domestic equilibrium price p^* of steel is determined by the intersection of domestic supply and demand, as illustrated in Figure 4.8. In the figure, consumer surplus is represented by the blue area, and producer surplus by the red area.

Figure 4.8: Small economy before opening to international trade

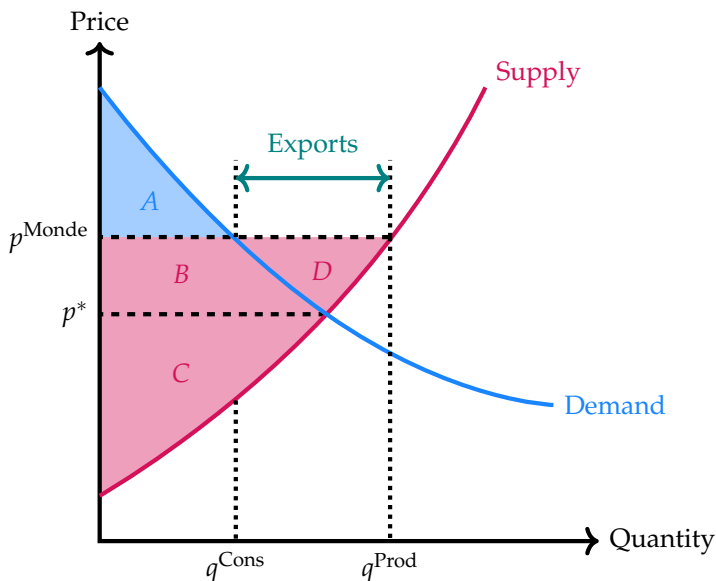


When the country opens its borders to international trade, anyone can buy or sell steel on the world market at the prevailing world price p^{World} . According to the law of price uniqueness, the domestic steel price must now equal p^{World} . Indeed, no domestic consumer would pay more than p^{World} for steel, as they could instead purchase it abroad at that lower

price. Similarly, no domestic producer would accept less than p^{World} , as they could sell their steel to foreign buyers at that higher price. Thus, the world price p^{World} acts simultaneously as a price ceiling and a price floor.

How does the domestic economy respond to this new price? The outcome depends explicitly on how p^{World} compares with the initial domestic equilibrium price p^* . If $p^{\text{World}} > p^*$, the domestic steel price rises. Consumers consequently reduce their quantity demanded to q^{Cons} , as illustrated in Figure 4.9. Domestic producers, however, increase their output to the level q^{Prod} . The difference between q^{Prod} and q^{Cons} represents the quantity of steel the country exports.

Figure 4.9: Small open economy with a world price above the autarky price



What is the effect of this price increase on surplus? Figure 4.9 illustrates that opening trade reduces consumer surplus in the steel market from $A + B$ to A . Consumers thus lose from the introduction of free trade. Producers, however, experience a significant increase in surplus—from area C to area $B + C + D$, and are therefore the beneficiaries of trade liberalization. Importantly, the gains to producers exceed the losses suffered by consumers. As a result, the economy's total surplus increases by area D , representing the national benefit from trade.

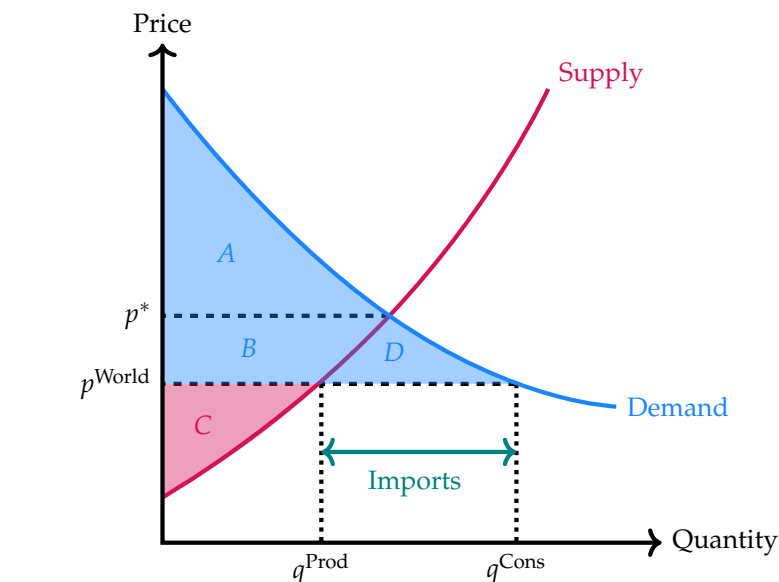
In principle, producers could transfer some of their gains to consumers to compensate them. For consumers to fully benefit, this redistribution would have to range between B and $B + D$. This possibility implies that protectionism is Pareto-inefficient: starting from autarky, it is feasible to

open trade and redistribute gains in a way that makes everyone strictly better off. In other words, opening trade increases the economy's total surplus, and not doing so would thus be inefficient. This reasoning explains economists' general opposition to trade barriers.

However, if such redistribution from winners to losers is not feasible in practice, protectionism could be Pareto-efficient. Indeed, starting from autarky, enhancing the welfare of those who would benefit from free trade would inevitably harm those who stand to lose.

When the world price p^{World} is lower than the autarky price p^* , the scenario is symmetric to that depicted in Figure 4.9. The domestic price of steel decreases, causing firms to reduce their production to the level q^{Prod} , as shown in Figure 4.10.⁴ Meanwhile, consumers respond by increasing their demand to the higher quantity q^{Cons} . The difference between q^{Cons} and q^{Prod} represents steel imports. In this case, consumers benefit from open borders, as their surplus rises from area A to $A + B + D$. Producers, by contrast, are harmed, experiencing a decline in surplus from $B + C$ to C . Yet, as before, the gains to consumers exceed producers' losses. It remains possible, at least in theory, for consumers to compensate producers fully, ensuring everyone prefers free trade to autarky.

Figure 4.10: Small open economy with a world price below the autarky price



⁴This adjustment can be interpreted in two ways: If all firms are identical and face rising marginal costs, each firm simply reduces its output. In the more realistic case where firms differ in productivity, opening to trade forces the highest-cost firms to exit the market.

This simple modeling approach provides a first approximation of who gains and who loses from international trade. When a country opens to trade, consumers benefit from lower prices of imported products, but domestic producers of those products are harmed. Conversely, domestic producers benefit when trade raises the prices of goods they export, but domestic consumers of those goods are adversely affected. As we saw clearly in the Ricardo model, open borders lower prices in sectors where the country is relatively less efficient and increase prices in sectors where the country holds a comparative advantage. Thus, sectors where the country has a comparative advantage benefit from free trade, while other sectors suffer. For example, France enjoys a comparative advantage in tourism, luxury goods, and aerospace; indeed, these sectors have expanded significantly with increased international openness. By contrast, French textile and metals industries have experienced substantial declines due to global competition.

4.2.2 Tariffs

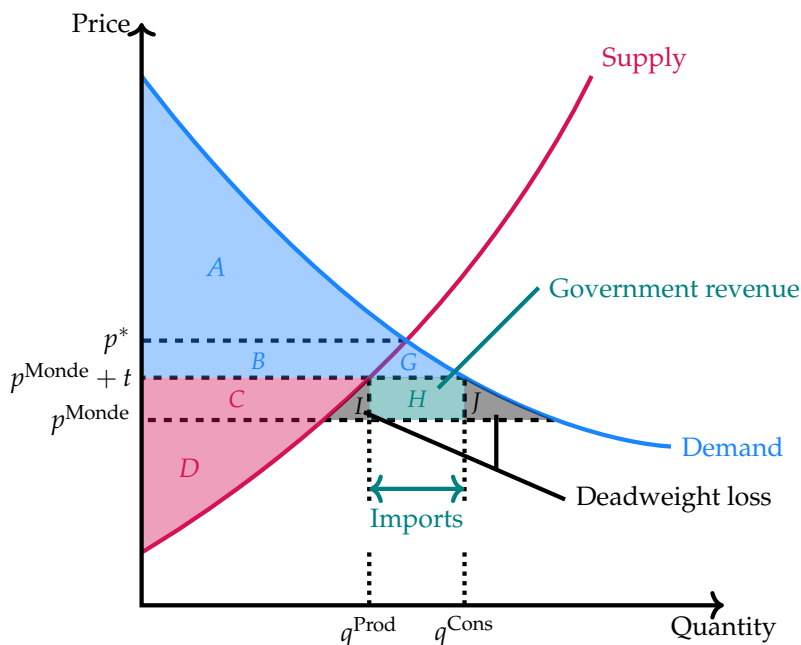
Domestic firms sometimes lobby their own governments to impose tariffs with the aim to protect themselves against increased foreign competition. What are the consequences of such policies? To address this, consider again the scenario in which producers are harmed by open trade, as illustrated in Figure 4.10. Now assume the government imposes a tariff t on each imported unit. The effective domestic price of imports then rises from p^{World} to $p^{\text{World}} + t$, as shown in Figure 4.11. This increase in domestic price leads to lower consumption and higher domestic production, thus reducing imports and moving the economy back in the direction of autarky. As intended by the producers' lobbying efforts, producer surplus increases by area C relative to the free-trade scenario. Consumers, however, face higher prices, causing their surplus to fall by $C + I + H + J$.

More precisely, Figure 4.11 clearly indicates the following effects after imposing the tariff:

- An amount C of consumer surplus transfers directly to producers.
- An amount H of consumer surplus transfers to the government in the form of tariff revenue. This revenue is represented by the green rectangle and equals the tariff t multiplied by the level of imports, $(q^{\text{Cons}} - q^{\text{Prod}})$.
- The amounts $I + J$ of lost consumer surplus are neither transferred to producers nor to the government; these losses represent the tariff's **deadweight loss**.

If the tariff is sufficiently large, the domestic price after opening the borders remains unchanged from the autarky level. This happens specifically when t is such that $p^{\text{World}} + t$ is at least p^* . In this case, imports fall to zero, government revenue from tariffs also becomes zero, and the deadweight loss exactly equals the entire unrealized gain from free trade.

Figure 4.11: Imposing a Tariff



The presence of this deadweight loss once again demonstrates the inefficiency of protectionist measures. Our analysis shows that the optimal policy for a small economy is to open its borders *unilaterally*.

The China Shock

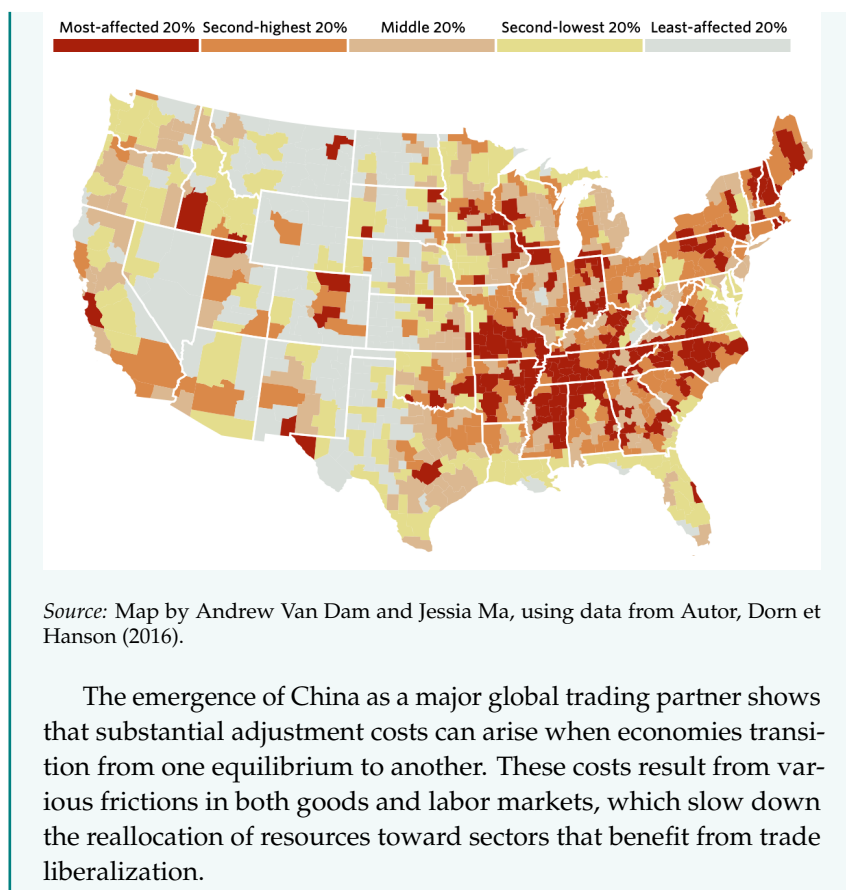
One of the most significant economic developments in recent decades is China's remarkable growth. Between 1990 and 2020—a single generation—China's standard of living increased nearly ten-fold. This rapid economic growth affected not only China's population, but also the entire world through international trade. Whereas China produced only 5% of global manufactured goods in 1990, its share rose to nearly 30% by 2020. Much of this increased produc-

tion is exported, particularly since China joined the World Trade Organization (WTO) in 2001.

What were the consequences of this dramatic rise in Chinese exports, known as the *China shock*? The impact was geographically concentrated. Industries tend to cluster in particular areas: textiles in northern France, high-tech in Silicon Valley, and so forth. The regions most heavily affected by Chinese competition were those specializing in industries in which China held a comparative advantage. In the United States, the *China shock* hit especially hard in the Southeast (see Figure 4.12), where manufacturing employment declined significantly.

In the long run, new jobs replaced those that disappeared: workers moved to other sectors or relocated to regions that benefited from increased trade with China. Unfortunately, this adjustment process took considerable time. Research by Autor, Dorn et Hanson (2016) shows that in the short run, this reallocation led to higher unemployment, reduced labor-force participation, and even increased mortality rates (particularly from overdoses and alcohol-related causes). The poorest workers—those less geographically mobile or less able to switch industries—faced especially severe impacts. It took over a decade for the economy to fully adjust. In the short run, the costs of transition were comparable in magnitude to the gains from trade liberalization. Over the longer term, however, the United States benefited significantly from opening trade with China. Even after accounting for transitional costs, roughly 95% of the U.S. population ultimately gained from the *China shock*.

Figure 4.12: Exposure of U.S. Commuting Zones to Chinese Exports



4.3 Other Effects of Free Trade

So far, we have introduced two models highlighting the main forces driving international trade and analyzed their implications for consumers and producers. Naturally, trade between countries has additional effects beyond those captured by our basic frameworks. What are these other effects, and what do they imply for trade policy? Let us briefly explore these questions.

4.3.1 Benefits of Free Trade

Access to More Varied Products. In industries characterized by substantial economies of scale, mass production is essential to minimize average manufacturing costs. For example, the costs associated with designing a car and setting up a production line are so large that the French market alone can sustain only a few automakers. Consequently, French

consumers have limited choices. By opening borders, many more car manufacturers can coexist in the global market. Increased global competition gives firms incentives to diversify their product offerings, expanding choices for consumers. Thus, French buyers no longer have to choose exclusively between Renault and Peugeot; they can also buy cars from BMW, Ford, Toyota, and many others.

In 1979, the American economist Paul Krugman was the first to formalize this type of international trade, which earned him the Nobel Prize in Economics in 2008. Krugman's "new" trade theory helps explain why a significant portion of international trade occurs between countries with similar productivity levels—a phenomenon Ricardo's theory of comparative advantage does not fully address. It sheds light on why Germany exports cars to France while France simultaneously exports cars to Germany. The reason is straightforward: German and French cars differ from each other, and consumers in each country have varied preferences.

Krugman's framework primarily accounts for "North-North" trade among similarly developed countries, whereas Ricardo's theory typically explains "North-South" trade between countries of differing productivity levels. Interestingly, since the 1980s, China's integration into the global economy has increased the prominence of trade based explicitly on comparative advantage. Despite their differences, both Ricardo's and Krugman's perspectives point to a similar economic policy implication: opening borders facilitates mutually beneficial trade.

Price Reductions through Intensified Competition. We will soon revisit a significant market failure: imperfect competition. When a small number of firms dominate a large market share, they often charge excessively high prices. International trade expands market size, intensifies competition, and thus lowers prices. Greater competition can also stimulate firms to innovate more actively.

When competition is inadequate, governments often attempt to regulate firms. However, as economist Milton Friedman famously noted, sometimes the most effective form of regulation is simply opening up markets to international competition.

Nevertheless, this reasoning does not always hold. In some cases, opening to international trade can weaken competition and worsen market failures. The digital industry provides a clear example. Trade liberalization has allowed global tech giants such as Google, Amazon, and Facebook to dominate international markets due to powerful network effects. The larger and more interconnected a firm's network becomes, the harder it is for competitors to displace it. China's decision to block these Western tech giants allowed domestic firms—such as Baidu, Alibaba,

and Tencent—to emerge, protecting Chinese users from dependence on U.S.-based digital platforms.

The Circulation of Ideas. Opening borders facilitates not only the exchange of goods but also the movement of people and ideas. Trade has long been a channel for spreading innovation. Historically, the Silk Road brought numerous Chinese technologies to Europe, including the compass and gunpowder. More recently, China’s rapid productivity growth over recent decades was closely linked to its integration into the global economy and the resulting transfer of knowledge and technology. Conversely, countries subject to international embargoes—such as Cuba or North Korea—find themselves isolated from the technological frontier and deprived of these productivity gains.

4.3.2 Dangers of Free Trade

Despite its significant benefits, free trade also involves certain dangers and undesirable effects that may justify restrictions on international commerce.

Infant Industry. Suppose Indonesia, thanks to the talent of its engineers, has the potential to develop a comparative advantage in the automotive sector. Initially, however, Indonesian factories are likely to produce lower-quality cars compared to established foreign competitors. It could take several years before the productivity gap with major multinational automakers closes. To enable Indonesia’s automotive sector to establish itself and mature, it may be beneficial to protect it during this initial “infant” stage. During this period, Indonesian consumers would have no choice but to buy domestically produced cars, which might initially offer poorer quality relative to their price. Over time, however, the quality of these cars would improve until the industry could compete internationally on equal terms.

This “infant industry” argument is especially relevant for developing countries and industries characterized by increasing returns to scale. However, it justifies only *temporary* trade barriers rather than permanent protectionism.

Challenges of Redistribution. In theory, a government could transfer resources from the winners to the losers of free trade so that everyone would benefit from opening borders, making protectionism Pareto-inefficient. In practice, implementing such redistribution proves challenging. Identifying precisely who gains and who loses from trade liberalization is often difficult. Furthermore, redistribution policies are politically sensitive and

challenging to enact. As a result, opening borders inevitably creates some losers who may not be fully compensated. Under these conditions, protectionism may no longer be inefficient. Policymakers then face a trade-off between economic prosperity and income inequality. Restricting trade reduces overall economic development but may increase incomes for the most vulnerable groups, such as low-skilled workers.

Environmental Protection. The European Union imposes strict environmental standards on firms operating within its borders. These stringent regulations inadvertently grant foreign countries a comparative advantage in more polluting industries. When pollution remains local, free trade poses minimal problems: each country can choose its environmental standards based on domestic preferences regarding pollution costs and the economic benefits of exporting environmentally damaging goods. However, when pollution has global consequences—such as greenhouse gas emissions—free trade significantly reduces the effectiveness of domestic environmental standards. Instead of protecting the global environment, strict standards may primarily disadvantage domestic industries, shifting greenhouse gas emissions to countries like China—a phenomenon known as “carbon leakage.” One partial solution is implementing a border carbon tax, though designing and enforcing such a measure can be challenging. As discussed further in the next chapter, trade policy could nevertheless play a key role in mitigating global climate change.

Geostrategic Dependence. Free trade encourages each country to specialize in specific sectors. Consequently, consumers import goods no longer produced domestically. Countries must therefore take care not to become overly reliant on a single trading partner for critical goods. For instance, Germany imported nearly half of its natural gas from Russia, leaving it highly vulnerable when the Russo-Ukrainian conflict escalated dramatically in 2022. Similarly, during the Covid-19 pandemic, countries capable of producing medicines and vaccines domestically could prioritize their own populations.

Nevertheless, achieving complete self-sufficiency in all strategic sectors is practically impossible. A less costly and more realistic solution involves diversifying supply sources, thus reducing vulnerability and preventing excessive strategic dependence.

Loss of National Sovereignty. There are two main types of trade barriers: tariff barriers—essentially taxes on imports, which we have previously discussed—and non-tariff or regulatory barriers designed to limit foreign competition. A historical example of regulatory barriers comes

from Jean-Baptiste Colbert's policies under Louis XIV, aimed at regulating the French textile market. To limit imports, highly arbitrary rules were placed on domestic textile manufacturers. For instance, fabrics produced in Châtillon had to contain exactly 1,216 threads, and those from Auxerre exactly 1,376. Fabrics failing to meet these precise standards could not be legally sold, and this effectively excluded foreign competitors. Today, free trade agreements are lengthy precisely because participating countries must harmonize numerous product regulations. The European Union's core achievement is its "Single Market," which establishes a unified set of rules enabling firms to freely access 450 million European consumers.

However, certain regulations reflect more than protectionist intentions—they embody specific national sovereignty decisions. For example, Europe has chosen an agricultural model less reliant on industrial techniques compared to the United States. Consequently, the EU imposes tighter restrictions on genetically modified organisms and prohibits beef raised using growth hormones and poultry washed with chlorine, both widespread practices in the U.S. Such differences illustrate the fundamental trade-off between deeper international economic integration and preserving national regulatory autonomy.

Trade and Economic Volatility

The theory of comparative advantage implies that opening borders encourages countries to specialize in certain industries. However, real-world economies constantly face various shocks—such as technological advances or shortages of raw materials—that affect some sectors more than others. One concern is that a highly specialized economy might become less resilient and more volatile in response to these economic fluctuations.

At the same time, trade provides a form of insurance. Beyond sector-specific shocks, entire national economies sometimes experience macroeconomic shocks, such as financial crises or political upheavals. In such situations, trade allows countries facing production shortfalls to import needed goods from abroad. Conversely, when domestic consumption declines, trade enables producers to find markets overseas. Thus, maintaining diversified trading partnerships can help stabilize economic activity.

Caselli, Koren, Lisicky et Tenreyro (2020) empirically studied these competing effects using data from 25 countries. They found that, on average, reductions in trade barriers from 1972 to 2007

reduced national income volatility by 36%. Thus, the stabilizing effect of trade diversification outweighed the volatility potentially introduced by sectoral specialization, making trade a stabilizing force overall.

4.3.3 Quantification of Free Trade Benefits

What are the measurable benefits of allowing goods to flow freely between regions? During the late 19th and early 20th centuries, transportation costs fell dramatically, primarily due to the expansion of railroads. What economic consequences did this innovation bring about? Ricardo's theory suggests increased specialization and associated gains from trade. Economists Arnaud Costinot and Dave Donaldson examined precisely this issue for U.S. agricultural production (Costinot et Donaldson, 2016). Using data from 1880 on a variety of agricultural products grown in different U.S. regions, they studied how these regions increasingly specialized in crops where they had a comparative advantage. Their research finds that improvements in transportation infrastructure since 1880 boosted U.S. agricultural output as significantly as technological advancements did. Thus, gains from domestic specialization were substantial.

Such productivity gains also translate into higher individual incomes. Using historical data on the expansion of the Indian railway network, Donaldson (2018) found that villages gaining rail access experienced notable increases in average income. These findings help explain why infrastructure development constitutes a major component of international aid.

These results clearly illustrate the economic benefits from trade within a country. But what about international trade specifically? Costinot et Rodríguez-Clare (2018) estimated the benefits the United States derives from international trade by asking: how much would Americans have to pay to maintain their current living standards if national borders were completely closed to trade? Their analysis indicates this cost would range from 2% to 8% of U.S. Gross Domestic Product—i.e., between 2% and 8% of all economic output produced annually. These international trade gains, though substantial, are considerably smaller than those obtained from domestic trade. This difference arises because the United States, given its large size, can produce a wide range of goods domestically. Indeed, American imports accounted for only 15% of GDP in 2018. For smaller countries like Belgium, whose imports represented 81% of GDP in 2018, gains from international trade are proportionally larger. Moreover,

these calculations do not even account for additional productivity benefits generated by the diffusion of ideas accompanying international trade.

4.3.4 Trade Deficits and Mercantilism

The protectionist policies implemented by Colbert were strongly influenced by mercantilism, an economic school of thought dominant from the 16th to the 18th century. According to mercantilist doctrine, a sound economic policy should seek to generate a surplus in the trade balance—maximizing exports while minimizing imports. Mercantilist policies frequently involved export subsidies and substantial tariffs.

Our analysis of international trade thus far has not addressed the question of trade balances directly, since Ricardo's model explicitly assumes that exports are the only means of financing imports. As a result, the trade balance is always perfectly balanced.

To examine trade surpluses and deficits explicitly, we must extend Ricardo's framework into a *dynamic* model of international trade. Such dynamic models form the basis of a specialized field called *international macroeconomics*, which studies the relationships between trade flows and international financial flows. We will explore this topic in Chapter 14. There, we will see that the central insight of the static approach—that exports are ultimately the only means to finance imports—still holds true in the dynamic setting. Therefore, the trade balance must remain balanced in the long run: surpluses in one period must inevitably be offset by deficits in future periods, and vice versa.

In the 18th century, philosophers such as David Hume and economists like Adam Smith and David Ricardo strongly opposed mercantilism. They argued that maintaining a persistently positive trade balance imposed high short-term costs and was unsustainable over the long term. These insights eventually led to mercantilism's decline and the emergence of the classical school of economics, from which much of contemporary economic thought originates.

Conclusion

In light of economic modeling and empirical studies, what can we ultimately say about the winners and losers from global trade? Is the widespread skepticism toward international trade justified? Are economists blind-sided when defending globalization?

Ricardo's core insight—that trade based on comparative advantage generates significant economic benefits—remains fundamentally sound. Yet this insight should only be taken as a first approximation. The picture

becomes more nuanced once we consider how different economic agents are affected. As shown explicitly by the small open economy model, opening up to trade inevitably disrupts domestic markets. Industries with a comparative advantage benefit from globalization, although domestic consumers in these sectors might legitimately complain about higher prices. Conversely, when an industry lacks comparative advantage, consumers happily enjoy reduced import prices, but domestic workers specialized in this industry lose their job and their skill becomes less valuable.

Overall, surplus analysis demonstrates that trade gains outweigh losses. This means that, in principle, free trade can therefore benefit everyone, provided that appropriate compensation mechanisms are in place. However, such redistribution policies are complex to implement. They are frequently neglected, and when they are not they primarily reflect political considerations, leading to potentially unfair outcomes. In theory, workers displaced from industries lacking comparative advantage can move toward better jobs in expanding sectors. In practice, however, this adjustment is challenging: acquiring new skills requires time and resources, and geographical relocation is costly, particularly when industries are regionally concentrated. Workers who possess greater adaptability, whether through education, mobility, or retraining opportunities, typically manage this transition successfully. In contrast, less adaptable workers often endure prolonged economic hardship. The resulting hardship can lead to political polarization and social tensions, fueling support for protectionist and mercantilist policies.

In the medium to long run, international trade remains a powerful driver of prosperity, economic growth, technological innovation, and even global peace. Yet, we cannot overlook the significant disruptions that accompany trade liberalization. Effectively mitigating these negative impacts and broadly sharing the benefits of trade demands a careful and nuanced policy approach. Addressing these complex issues remains one of the most significant economic and political challenges of our time. Economics, as a scientific discipline, contributes by clarifying public debates, informing policy decisions aimed at achieving widely shared prosperity, and helping societies navigate complex geopolitical realities.

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Exercises for Chapter 4

Exercise 4.1 Conscription (Military Draft)

Suppose a country has 50 skilled workers and 50 unskilled workers. A skilled worker can produce either 6 units of industrial goods or 2 units of national defense; an unskilled worker can produce either 2 units of industrial goods or 1 unit of national defense.

1. Show that unskilled workers have a comparative advantage in the production of defense.
2. Assume this country decides to produce 30 units of defense. Will the country have more industrial goods left for consumption if defense is produced by skilled workers or by unskilled workers? Is this result surprising? Explain.
3. Now suppose the government introduces conscription (mandatory military service) by randomly selecting 10 skilled and 10 unskilled workers for the production of defense, so that together they produce 30 units of defense. What is the resulting level of industrial goods output in this scenario?
4. An alternative to conscription is a voluntary army in which the state pays unskilled workers enough to persuade 30 of them to become soldiers. Compare the opportunity cost (in terms of industrial goods) of producing 30 units of defense under these two options: conscription versus a voluntary army. What about the budgetary cost?
5. Is the model used in this exercise “useless” or “silly”? Discuss.

Exercise 4.2 The Ricardian Model

Let us return to the Ricardian model from the chapter, where the number of labor hours required to produce cars and boats is given by:

	Car	Boat
Germany	50	100
France	125	125

Each country has 500 hours of labor. We assume that initially the global (Franco-German) demand is such that in an open economy, the relative price of boats lies between 1 and 2, i.e.

$$\frac{p_{\text{Boat}}^{\text{Inter}}}{p_{\text{Car}}^{\text{Inter}}} \in (1, 2).$$

1. **Hourly Wages and Specialization.** Calculate the hourly wage of a German worker and of a French worker in each sector. Which sectors will they choose to specialize in? Do German workers earn more than French workers?
2. **Constructing Figure 4.6.** Explain how Figure 4.6 is constructed.
3. **Relative Price of Boats Equal to 2.** Suppose there is a strong increase in demand for boats such that

$$\frac{p_{\text{Boat}}^{\text{Inter}}}{p_{\text{Car}}^{\text{Inter}}} = 2.$$

What is the impact of this change on each of the two countries? Provide an intuitive explanation.

4. **Doubling Germany's Productivity in Both Sectors.** Again assume that the relative price of boats initially lies between 1 and 2, i.e.

$$\frac{p_{\text{Boat}}^{\text{Inter}}}{p_{\text{Car}}^{\text{Inter}}} \in (1, 2),$$

and suddenly Germany experiences a doubling of its productivity in both car and boat production. What is the impact of this change on the French economy? Give an intuitive explanation.

5. **Doubling Germany's Productivity in Boat Building Only.** Now assume that Germany's productivity doubles *only* in boat building. What is the impact of this change on the French economy? Provide an intuitive explanation.
6. **Beyond the Ricardian Model: A Few German Workers in France.** Let us now go beyond the Ricardian model by assuming there is a small number of German workers living in France (with hourly

productivities as given in the table above). Are these workers in favor of opening the borders? Explain.

7. **Winners and Losers?** Why does free trade never create losers in the Ricardian model?

Exercise 4.3 Compensations and Efficiency

1. Does the existence of losers from free trade justify protectionism? Which assumption alters your conclusion? Explain.
2. Is free trade without compensation Pareto-efficient? Explain.
3. Should the losers from free trade be compensated?